

ZEPTO SQL DATA ANALYSIS

● CREATION OF DATABASE

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1 • CREATE DATABASE ecom;
```

● CREATION OF TABLE

```
1  -- TABLE CREATION
2
3 •  USE ecom;
4 •  DROP TABLE IF EXISTS zepto;
5 •  CREATE TABLE zepto(
6      id INT AUTO_INCREMENT PRIMARY KEY,
7      category VARCHAR(100),
8      name VARCHAR(200) NOT NULL,
9      mrp NUMERIC(8,2),
10     discountPercent NUMERIC(5,2),
11     availableQuantity INT,
12     discountedSellingPrice NUMERIC(8,2),
13     weightInGms INT,
14     outOfStock BOOLEAN,
15     quantity INT
16 );
17
18 •  SELECT * from zepto;
19
20
```

id: Unique identifier for each product entry (Synthetic Primary Key)

name: Product name as it appears on the app

category: Product category like Fruits, Snacks, Beverages, etc.

mrp: Maximum Retail Price (originally in paise, converted to ₹)

discountPercent: Discount applied on MRP

discountedSellingPrice: Final price after discount (also converted to ₹)

availableQuantity: Units available in inventory

weightInGms: Product weight in grams

outOfStock: Boolean flag indicating stock availability

quantity: Number of units per package (mixed with grams for loose produce)

● Data Exploration

Counted the total number of records in the datasets

Viewed a sample of the data set to understand structure and content

Checked for null values across all columns

Identified distinct product categories available in the data set

Compared in-stock vs out-of-stock product counts

Detected products present multiple times, representing different ids

● DATA CLEANING

Identified and removed rows where MRP or discounted selling price was zero.

Converted mrp and discountedSellingPrice from paise to rupees for consistency and readability

● BUSINESS INSIGHTS

- a. Found top 10 best-value products based on discount percentage
- b. Identified high-MRP products that are currently out of stock
- c. Estimated potential revenue for each product category
- d. Filtered expensive products (MRP > ₹500) with minimal discount
- e. Ranked top 5 categories offering highest average discounts
- f. Calculated price per gram to identify value-for-money products
- g. Grouped products based on weight into Low, Medium, and Bulk categories
- h. Measured total inventory weight per product category

● SCREENSHOTS-

```
3  -- Q1. Find the top 10 best-value products based on the discount percentage.
4
5  • select DISTINCT name, mrp, discountPercent
6     from zepto
7     order by discountPercent DESC
8     LIMIT 10;
9
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	name	mrp	discountPercent
▶	Dukes Waffy Chocolate Wafers	45.00	51.00
	Dukes Waffy Orange Wafers	45.00	51.00
	Dukes Waffy Strawberry Wafers	45.00	51.00
	Ceres Foods Fish Mustard Instant Liquid Masala	220.00	50.00
	Ceres Foods Laal Maas Instant Liquid Masala	220.00	50.00
	Ceres Foods Nalli Nihari Instant Liquid Masala	220.00	50.00
	Chef's Basket Durum Wheat Elbow Pasta	160.00	50.00
	Chef's Basket Durum Wheat Fusilli Pasta	160.00	50.00
	Chef's Basket Durum Wheat Penne Pasta	160.00	50.00
	Dukes Waffy Chocolate Wafer Rolls	150.00	50.00

```
10  -- Q2.What are the Products with High MRP but Out of Stock?
11
12  • SELECT distinct name, mrp
13     FROM ZEPTO
14     where outOfStock= 1 AND mrp>400
15     order by mrp desc;
16
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	name	mrp	discountPercent
	Dukes Waffy Chocolate Wafers	45.00	51.00
	Dukes Waffy Orange Wafers	45.00	51.00
	Dukes Waffy Strawberry Wafers	45.00	51.00
	Ceres Foods Fish Mustard Instant Liquid Masala	220.00	50.00
	Ceres Foods Laal Maas Instant Liquid Masala	220.00	50.00
	Ceres Foods Nalli Nihari Instant Liquid Masala	220.00	50.00
	Chef's Basket Durum Wheat Elbow Pasta	160.00	50.00
	Chef's Basket Durum Wheat Fusilli Pasta	160.00	50.00
	Chef's Basket Durum Wheat Penne Pasta	160.00	50.00
	Dukes Waffy Chocolate Wafer Rolls	150.00	50.00

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17  -- Q3.Calculate Estimated Revenue for each category
18
19  •  select category,
20      sum(discountedSellingPrice * availableQuantity) as total_revenue
21      from zepto
22      group by category
23      order by total_revenue;
24

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	name	mrp
▶	Patanjali Cow's Ghee	565.00

```

25  -- Q4. Find all products where MRP is greater than ₹500 and discount is less than 10%.
26
27  •  select distinct name, mrp, discountPercent
28      from zepto
29      where mrp>500 AND discountPercent<10
30      order by mrp desc, discountPercent desc;
31

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	category	total_revenue
▶	Fruits & Vegetables	10846.00
▶	Meats, Fish & Eggs	20693.00
▶	Biscuits	25007.60
▶	Dairy, Bread & Batter	55051.00
▶	Beverages	55051.00
▶	Health & Hygiene	64180.00
▶	Home & Cleaning	122661.00
▶	Packaged Food	224385.00
▶	Ice Cream & Desserts	224385.00
▶	Chocolates & Candies	224385.00
▶	Personal Care	270849.00
▶	Paan Corner	270849.00

```

32 -- Q5. Identify the top 5 categories offering the highest average discount percentage.
33
34 • select category,
35    ROUND(avg(discountPercent),3) AS avg_discount
36 from zepto
37 group by category
38 order by avg_discount desc
39 LIMIT 5;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

name	mrp	discountPercent
Dhara Kachi Ghani Mustard Oil Jar	1250.00	8.00
Saffola Gold (Jar)	1240.00	0.00
Dhara Filtered Groundnut Oil (Jar)	1050.00	1.00
Fortune Rice Bran Health Oil (Jar)	1050.00	1.00
Dhara Filtered Groundnut Oil (Jar)	1050.00	0.00
Fortune Soyabean Oil	1005.00	0.00
Fortune Sunlite Refined Sunflower (Jar)	925.00	0.00
Surf Excel Matic Powder Front Load	810.00	7.00
Surf Excel Matic Top Load	720.00	9.00
Pedigree Puppy Dry Dog Food Food Chicken & Milk	690.00	6.00
Pedigree Dog Food Adult Meat & Rice	660.00	7.00

```

41 -- Q6. Find the price per gram for products above 100g and sort by best value.
42
43 • select distinct name, weightInGms, discountedSellingPrice,
44    ROUND(discountedSellingPrice/weightInGms,2) AS price_per_gram
45 from zepto
46 where weightInGms>=100
47 order by price_per_gram ;
48

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content:  | Fetch rows: 

category	avg_discount
Fruits & Vegetables	15.462
Meats, Fish & Eggs	11.032
Chocolates & Candies	8.325
Ice Cream & Desserts	8.325
Packaged Food	8.325

```

50 -- Q7.Group the products into categories like Low, Medium, Bulk.
51
52 • select distinct name, weightInGms,
53 case
54     when weightInGms<1000 then 'low'
55     when weightInGms<4000 then 'medium'
56     else 'bulk'
57 END AS weight_category
58 from zepto;

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	name	weightInGms	weight_category			
▶	Onion	1000	medium			
	Tomato Hybrid	1000	medium			
	Tender Coconut	58	low			
	Coriander Leaves	100	low			
	Ladies Finger	250	low			
	Potato	1000	medium			
	Lemon	200	low			
	Watermelon	58	low			
	Capsicum Green	250	low			

```

61 -- Q8.What is the Total Inventory Weight Per Category
62
63 • select distinct category,
64     ROUND(sum( weightInGms* availableQuantity),2) AS total_weight
65 from zepto
66 group by category
67 order by total_weight;
68
69

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	name	weightInGms	weight_category			
▶	Onion	1000	medium			
	Tomato Hybrid	1000	medium			
	Tender Coconut	58	low			
	Coriander Leaves	100	low			
	Ladies Finger	250	low			
	Potato	1000	medium			
	Lemon	200	low			
	Watermelon	58	low			
	Capsicum Green	250	low			
	Chilli Green	100	low			
	Banana Robusta	348	low			